RAHUL MITRA

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EDUCATION

National University of Singapore **Aug 2023 – Jul 2027**
Bachelor of Engineering, Industrial and Systems Engineering (GPA: 4.37/5.00)

- Pursuing a Second Major in Computer Science and a Minor in Mathematics.
- Awarded Certificate of Outstanding Performance in 3D Computer Vision (CS4277): top student in a class of 24, NUS School of Computing.

WORK EXPERIENCE

Warehouse AI & Operations Research Engineer (Part-Time), STMicroelectronics (NUS Capstone) **Aug 2026 – Present**

- Architected a layout-agnostic AI put-away engine to run in any STMicroelectronics warehouse worldwide, ingesting each site's shelf coordinates, dimensions, and operating rules so one model adapts to any floor plan.

Global Operational AI Systems & Data Engineer (Part-Time), Abbott Laboratories (Contract) **Jun 2026 – Present**

- Develop a chat-based spray-dryer fault-detection tool validated across 5+ plants worldwide, cutting critical-failure diagnosis from days or months of manual raw-signal analysis to within an hour, flagging inconclusive signals rather than false positives.
- Assemble a 5-stage offline process-intelligence platform for globally distributed spray-drying plants: GPU OCR and layout parsing, recursive chunking, ONNX embeddings, a Chroma vector store, and LangChain over a locally hosted Gemma model.

Platform & Solutions Engineer (Sparks Citizen Developer), People's Association **Sep 2025 – Sep 2026**

- Led most software development and deployment of a digital garden-plot balloting platform across admin and public Next.js frontends, a NestJS backend, and GIS mapping, now live in Tengah with national rollout planned.

Data Analytics & Product Management Intern, Abbott Laboratories (Converted to Contract) **Jan 2026 – Jun 2026**

- Developed a SimPy discrete-event digital twin and OR-Tools CP-SAT solver for hybrid flow-shop scheduling, then compared 3 approaches (heuristic, MIP, and genetic algorithm) to turn production objectives and constraints into decision-ready schedules.
- Trained regional engineering teams across Asia and Europe on repeatable AI workflows spanning 7 areas: prompting, model selection, benchmarks, cost awareness, agents, validation, and safeguards.

RESEARCH EXPERIENCE

Computer Vision Researcher, BlendED x Amazon Robotics (AI+X Global Challenge) **Jul 2026 – Oct 2026**

- Built vision systems turning raw camera feeds into perception-ready frames across 3 workstreams: camera ISP, super-resolution, and image restoration under real-world blur.
- Reproduced a published UNet super-resolution model in PyTorch from broken author code, recovering its undocumented evaluation protocol and training 12 models across 3 scales, 2 datasets, and 2 losses, matching reported results at 8x and landing within 0.78 dB at 2x and 4x.

PROJECTS

Team Lead & Finalist, Swarmline: SG Defence Tech Hackathon (1,300+ applicants) **Sep 2026**

- Directed a 5-person team to architect decentralised drone-swarm coordination on correlated random-walk search, gradient-flow dispersion, delay-tolerant gossip, cross-inhibition consensus, and response-threshold task allocation.

Finalist, Volt Pulse SG: SMU Hack For Cities (Top 8 Finalist of 50+ teams) **Jan 2026**

- Integrated SEALION embeddings, Supabase vector search, RRF routing, and scheduler-agent planning into an autonomous loop managing electricity and utility savings across multiple HDB households at once.

Team Lead, Maritime Deficiency Severity Forecasting: Maritime Hackathon **Jun 2025**

- Led model training for a maritime-inspection NLP system, fine-tuning a 109M-parameter BERT with DNN layers and gradient clipping on the ASPIRE 2A supercomputer.

LEADERSHIP AND ACTIVITIES

Elderly Care Volunteer, NTUC Health Active Ageing Centre **Sep 2025 – Sep 2026**

- Coordinated weekly senior-engagement activities, including home visits to check health, hear life stories, and help with everyday tasks, building empathy for accessible, community-focused product design.

SKILLS AND CERTIFICATIONS

Technical: Python, PyTorch, BERT, DNN, RAG, AI agents, GPT-4, Gemini, Claude Code, ChatGPT, Codex, ComfyUI, SEALION, ONNX, Chroma, local LLM serving (llama.cpp), vector search, model evaluation, NLP; vision: camera ISP, super-resolution, image restoration, camera models and calibration, fundamental and essential matrices, epipolar geometry, absolute pose estimation, three-view geometry, structure-from-motion (SfM) with bundle adjustment, two- and multi-view stereo, generalised camera models; HPC: ASPIRE 2A, CUDA; deep RL (PPO, A2C, DDPG, DQN)
Certifications: NVIDIA Disaster Risk Monitoring with Satellite Imagery